

Radians-problems involving area 5ii

Subject & topics: Maths Stage & difficulty: A Level P1

Figure 1 shows a sector OAB of a circle, centre O and radius 8 cm. The angle AOB is 46° .

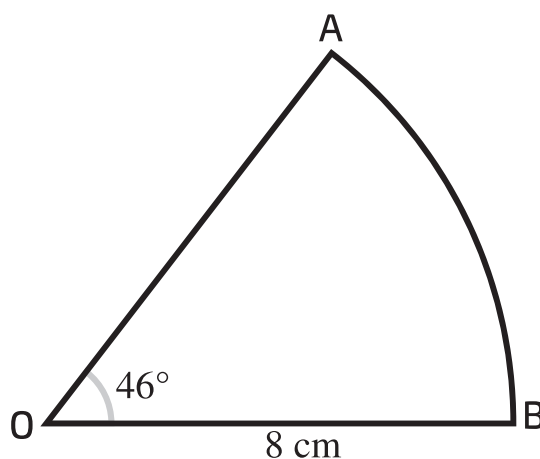


Figure 1: Sector AOB.

Part A

Convert angle to radians

Express 46° in radians, correct to 3 significant figures.

Part B

Arc length

Find the length of the arc AB.

Part C

Area of sector

Find the area of the sector OAB.

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Question deck:

STEM SMART Double Maths 24 - Radians, Trigonometric

Reciprocals & t-Formulae

Radians-problems involving area 1ii

Subject & topics: Maths Stage & difficulty: A Level P1

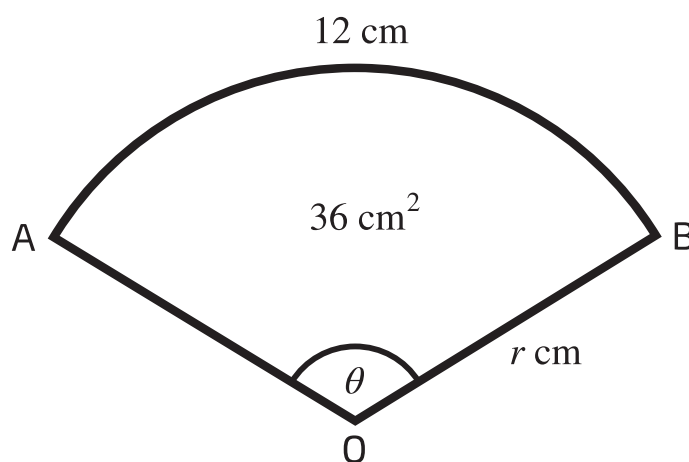


Figure 1: The sector OAB.

A sector OAB of a circle of radius r cm has angle θ radians. The length of the arc of the sector is 12 cm and the area of the sector is 36 cm^2 (see Figure 1).

Part A

First equation

By considering the length of the arc of the sector, write down an equation involving r and θ , where one side of the equation is a numerical constant.

The following symbols may be useful: r , θ

Part B

Second equation

By considering the area of the sector, write down another equation involving r and θ , where one side of the equation is a numerical constant.

The following symbols may be useful: r , θ

Part C

Values of r and θ

Hence show that $r = 6$ cm and find the value of θ .

Part D

Area of segment

Find the area of the segment bounded by the arc AB and the chord AB. Give your answer to 3 sf.

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Reciprocals & t-Formulae

Radians and Trig Functions 1i

Subject & topics: Maths **Stage & difficulty:** A Level P2

A curve has equation $y = \sin(ax)$, where a is a positive constant and x is in radians.

Part A Period

State the period of $y = \sin(ax)$, giving your answer in an exact form in terms of a .

The following symbols may be useful: a , π

Part B $\sin(ax) = k$

The two smallest positive solutions of $\sin(ax) = k$, where k is a positive constant, are $x = \frac{1}{5}\pi$ and $x = \frac{2}{5}\pi$.

Find the exact values of a and k .

$$a = \boxed{}$$

$$k = \boxed{} \sqrt{\boxed{}}$$

Part C

$$\sin(ax) = \sqrt{3} \cos(ax)$$

Given instead that $\sin(ax) = \sqrt{3} \cos(ax)$, find the two smallest positive solutions for x , giving your answers in an exact form.

Enter your answers in order from lowest value of x to highest.

$$x = \text{[input box]} \frac{\pi}{a} \text{ (lowest value)}$$

$$x = \text{[input box]} \frac{\pi}{a} \text{ (highest value)}$$

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Reciprocals & t-Formulae

Small Angle Approximations 3ii

Subject & topics: Maths **Stage & difficulty:** A Level P1

Part A

Expression

Use the small angle approximation to write an approximate expression for $(\cos \theta + 5)(1 + \sin(2\theta))$ in powers of θ up to θ^2 .

The following symbols may be useful: theta

Part B

Approximation

Use your answer from the above part to work out an approximate value for $(\cos \theta + 5)(1 + \sin(2\theta))$ when $\theta = 0.075$. Give your answer to 3 significant figures.

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Reciprocals & t-Formulae**

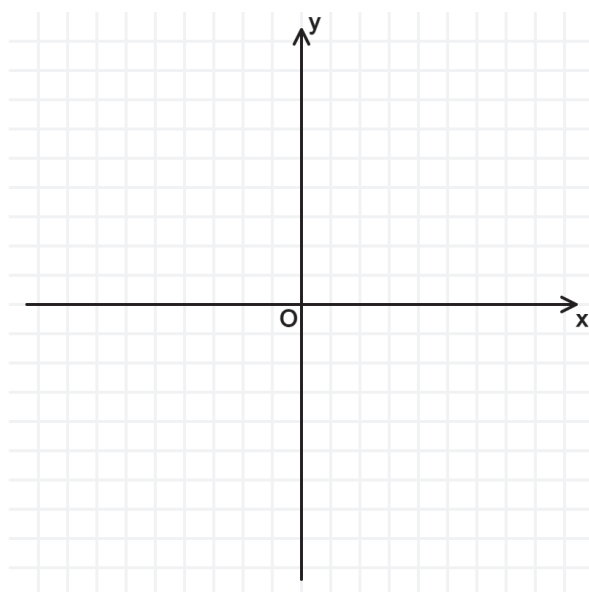
Sketching Reciprocal Trigonometric Functions

Subject & topics: Maths | Functions | Graph Sketching **Stage & difficulty:** A Level P2

Part A

Sketch $2 \sec x + 2$

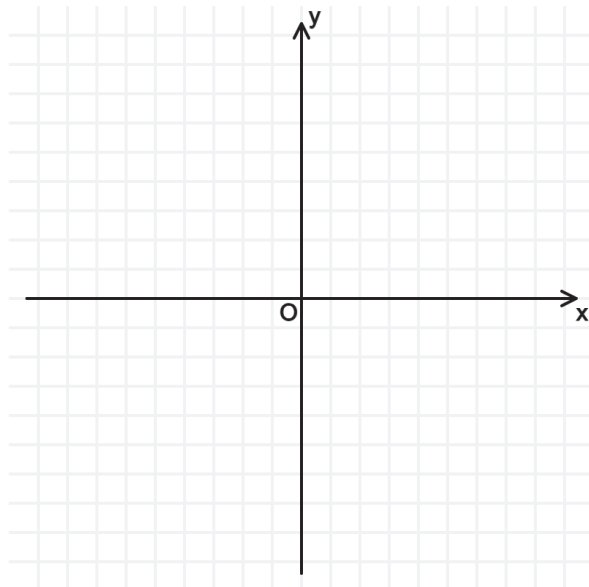
Sketch the graph of $y = 2 \sec x + 2$ in the interval $0^\circ \leq x \leq 360^\circ$.



Part B

Sketch $\cot\left(x + \frac{\pi}{4}\right) - 1$

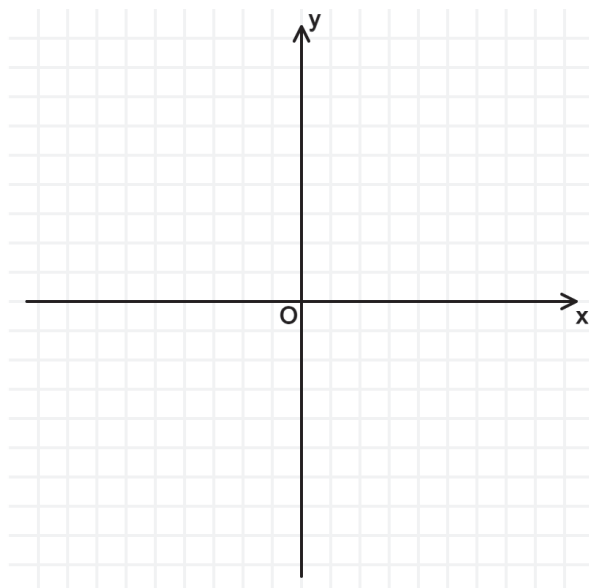
Sketch the graph of $y = \cot\left(x + \frac{\pi}{4}\right) - 1$ in the interval $0 \leq x \leq 2\pi$.



Part C

Sketch $-\operatorname{cosec} \frac{x}{2} - 3$

Sketch the graph of $y = -\operatorname{cosec} \frac{x}{2} - 3$ in the interval $-2\pi \leq x \leq 2\pi$.



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Question deck:

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Reciprocals & t-Formulae

Advanced Trig Identities 2ii

Subject & topics: Maths **Stage & difficulty:** A Level P2

Part A

$$2 \tan^2 \theta - \frac{1}{\cos \theta}$$

Express $2 \tan^2 \theta - \frac{1}{\cos \theta}$ in terms of $\sec \theta$.

The following symbols may be useful: $\sec()$, θ

Part B

Solve

Hence solve, for $0^\circ < \theta < 360^\circ$, the equation

$$2 \tan^2 \theta - \frac{1}{\cos \theta} = 4$$

Enter your answers in order from lowest value of θ to highest. Give inexact answers to 1 dp.

(lowest value)

(highest value)

Sketching Inverse Trigonometric Functions

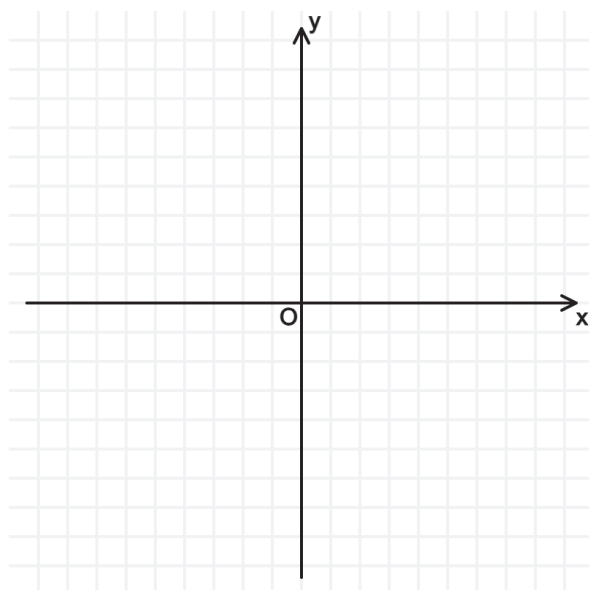
Subject & topics: Maths | Functions | Graph Sketching

Stage & difficulty: A Level P2

Part A

Sketch $\arcsin \frac{x}{3} + \frac{\pi}{2}$

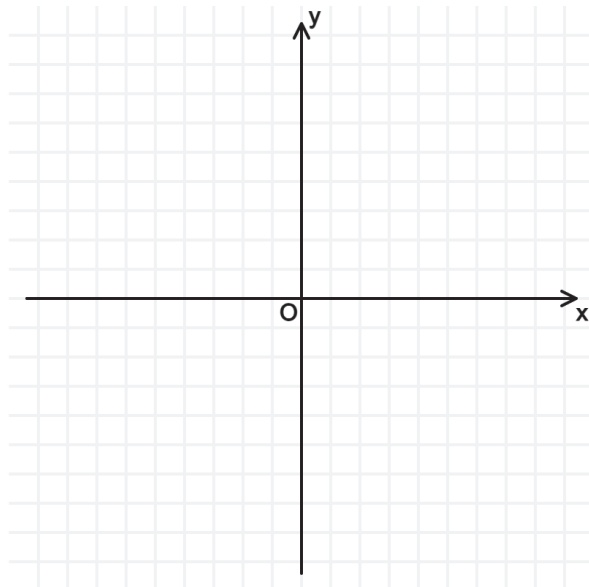
Sketch the graph of $y = \arcsin \frac{x}{3} + \frac{\pi}{2}$.



Part B

Sketch $2 \arctan(x - 1)$

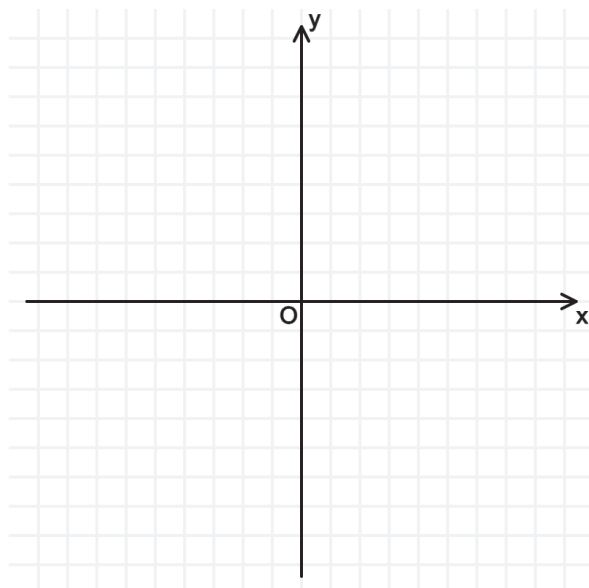
Sketch the graph of $y = 2 \arctan(x - 1)$.



Part C

Sketch $\arccos\left(-\frac{x}{2} + \frac{1}{2}\right) - \frac{\pi}{3}$

Sketch the graph of $y = \arccos\left(-\frac{x}{2} + \frac{1}{2}\right) - \frac{\pi}{3}$.



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Reciprocals & t-Formulae

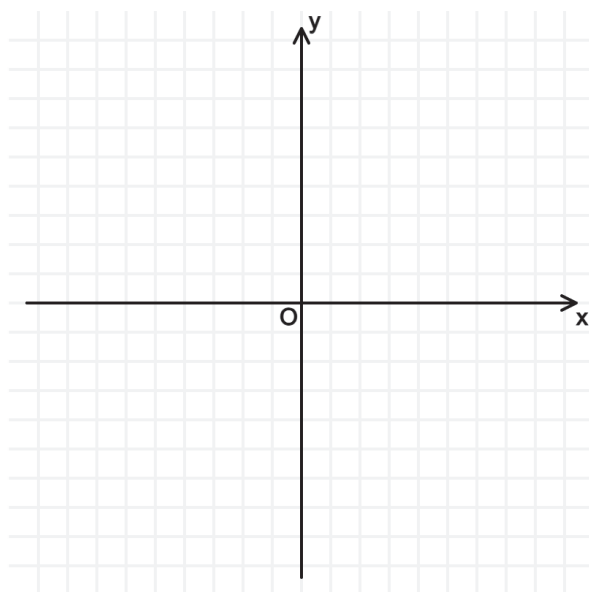
Functions: Reciprocal Trig 1i

Subject & topics: Maths Stage & difficulty: A Level P2

Part A

Sketch

Sketch the graph of $y = \operatorname{cosec} x$ for $0 < x < 4\pi$.



Part B

β in terms of α

It is given that $\operatorname{cosec} \alpha = \operatorname{cosec} \beta$, where $\frac{1}{2}\pi < \alpha < \pi$ and $2\pi < \beta < \frac{5}{2}\pi$. By using your sketch, or otherwise, express β in terms of α .

The following symbols may be useful: alpha, beta, pi

Part C

Double angle $\tan$ properties

Write down the identity giving $\tan 2\theta$ in terms of $\tan \theta$.

The following symbols may be useful: $\cos()$, $\sin()$, $\tan()$, θ

Part D

$\tan \phi \cot 2\phi \tan 4\phi$

Given that $\cot \phi = 4$, find the exact value of $\tan \phi \cot 2\phi \tan 4\phi$.

The following symbols may be useful: ϕ

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Reciprocals & t-Formulae

t-Formulae Substitution

Subject & topics: Maths | Geometry | Trigonometry **Stage & difficulty:** Further A P3

Part A

Substitution

Using the substitution $t = \tan \frac{\theta}{2}$, write the equation $2 \cos \theta - 5 \sin \theta = 2 - 5 \tan \theta$ in the form $f(t) = 0$ where $f(t)$ is a polynomial with integer coefficients and degree 4.

The following symbols may be useful: $\cos()$, $\operatorname{cosec}()$, $\cot()$, $\sec()$, $\sin()$, t , $\tan()$

Part B

Solutions

Hence find all the solutions to the equation $2 \cos \theta - 5 \sin \theta = 2 - 5 \tan \theta$ in the range $0 \leq \theta < 2\pi$. Where necessary, give irrational solutions to 3 sf.

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